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Reader-facing sound-change pilot

This assembled pilot PDF is for checking the first reader-facing rewrite layer of the sound-change half.

Included sections

1. Velar palatalization before front vowels
2. The Old English i-umlaut and its West Saxon right edge
3. Nasal dissimilation

Velar palatalization before front vowels

Internal rule and chronology card: OEVelarPalatalization (SC052).

1. Historical discussion

Older German scholarship places this change inside a broad early palatal phase. Luick opens the relevant sequence with the heading “**Frühe Verschiebungen in palataler Richtung**” and immediately treats English k and g before bright vowels as part of that field (Luick 1914–1940,

sect. 168). The older tradition is not wrong to frame the matter broadly: plain velars, sk, and later palatal effects do belong to one neighborhood.

Newer English-language scholarship is more explicit about the sequencing inside that neighborhood. Ringe and Taylor write that **“After initial velars and *sk had been palatalized” later West-Saxon diphthongization follows, which makes plain velar palatalization a real earlier consonantal stage rather than a mere side note (Ringe and Taylor 2014, sect. 6.5.1). Campbell likewise distinguishes plain velars from the broader sk complex when he notes that “[sk] is more prone to palatalization and assibilation than [k]”** (Campbell 1959, sect. 440).

The result is a familiar historical picture: the older tradition gives the large palatal field, while the newer English handbooks make it easier to isolate plain velar palatalization as a distinct step within that field.

2. Comparison of the traditions

The German and English traditions agree on the phenomenon but weight it differently.

Luick’s prose is large-scale and architectural. It describes an early movement toward palatal articulation and then places later vowel changes to the right of that region (Luick 1914–1940, sects 168–183). Campbell, Hogg, and especially Ringe and Taylor are more explicit about internal differentiation: plain velars, sk, and later front-mutation material are related, but they are not identical processes (Campbell 1959, sect. 170; Hogg 1992, 106–7, 111–14; Ringe and Taylor 2014, sects 6.4.1, 6.5.1, 6.6.1–6.6.4).

That comparison matters for the present chapter. The change is substantial enough to deserve its own reader-facing section, but it should still be explained as part of the larger palatalization-to-umlaut corridor rather than as a self-contained sound law detached from its neighbors.

3. Formalization in the present project

The present implementation expresses the change with one helper definition for plain k and a second block for plain g:

```
define OEVelarPalatalizationKFront [
  {*k} -> {*tʃ} || .#. _ EnglishStarFrontVowel,
  {*k} -> {*tʃ} || _ [{*i} | {*ī}],
  {*k} -> {*tʃ} || _ {*í},
  {*k} -> {*tʃ} || [{*i} | {*ī}] _ EnglishStarFrontVowel,
  {*k} -> {*tʃ} || {*í} _ EnglishStarFrontVowel,
  {*k} -> {*tʃ} || [{*i} | {*ī}] _ .#. ,
  {*k} -> {*tʃ} || {*í} _ .#.
] .o. [
  {*k} {*k} -> {*tʃ} {*tʃ} || _ {*j}
] .o. [
  {*k} -> {*tʃ} || _ {*j}
] ;

define OEVelarPalatalization [
  OEVelarPalatalizationKFront
] .o. [
  {*g} -> {*dʒ} || _ EnglishStarFrontVowel,
  {*g} -> {*dʒ} || EnglishStarFrontVowel _ .#. ,
  {*g} -> {*dʒ} || EnglishStarFrontVowel _ EnglishStarFrontVowel,
  {*g} -> {*dʒ} || EnglishStarFrontVowel _ [EnglishStarConsonant - {*j}],
  {*g} {*g} -> {*dʒ} {*dʒ} || _ {*j}
] .o. [
  {*g} -> {*dʒ} || _ {*j}
] ;
```

In prose, the rule does exactly what the title suggests: it palatalizes plain *k* and *g* in front-vocalic and *j*-adjacent environments. The point of writing it as a separate rule is not to deny the larger palatal field, but to make the relative order of plain-velar palatalization, *sk*-palatalization, and umlautal developments testable.

4. Chronological placement

The order-sensitivity work gives this change a concrete local chronology.

Placed too early, before Sievers-law syncope, it breaks the derivation of *stretch*. With PGmc **strákkijanā* in the wrong order, the model produces *strecċan* instead of the expected Old English *streċċan* [Germanic/docs/sound_changes/order_tests/chronology_cards/SC052-oe-velar-palatalization.md].

Placed too late, after *i*-umlaut, it over-palatalizes forms such as *cow* and *lung*. The chronology card records that PGmc **kūi* then yields *ċy* instead of expected *cȳ*, and PGmc **lúnganjō* yields *lun-ġen* instead of expected *lungen* [Germanic/docs/sound_changes/order_tests/chronology_cards/SC052-oe-velar-palatalization.md].

That is the reader-facing reason for the rule's present position: it must come after the syncope that prepares forms like *stretch*, but before the umlautal stage that would otherwise create the wrong palatalized outputs in *cow* and *lung*.

5. Consequences for reconstructed forms

Once the rule is in place, plain velars before front vowels and *j* no longer remain plain. They become the palatal outcomes that later chapters presuppose. That matters not only for dictionary-like forms such as *cild* or *dæg*, but also for the broader relation between consonantal palatalization and later vowel-fronting processes (Campbell 1959, sect. 170; Ringe and Taylor 2014, sect. 6.4.1).

In other words, the rule is consequential because it creates the consonantal environment inherited by the umlaut chapter. Without it, later reconstructed forms are not merely shifted in date; they come out with the wrong consonant quality.

6. Remaining cautions

This section should remain narrow.

It is not a chapter on every Old English palatal development. *sk* belongs to a related but not identical part of the tradition, and the later umlaut chapter is still a different historical problem. Nor should the left-hand relation to Sievers-law syncope be inflated into a larger joint chapter: the *stretch* evidence shows a real dependency, but it does not turn the feeder process into a coequal historical unit.

The Old English i-umlaut and its West Saxon right edge

Internal rules and chronology cards: OEIUmlaut (SC055) and OEWsPalatalDiphthongization (SC056).

1. Historical discussion

For the older German tradition, the center of this chapter is unmistakable. Luick writes: “**Der wichtigste Fall von palataler Beeinflussung ... war die Veränderung der urenglischen Vokale durch i oder j der Folgesilbe**” (Luick 1914–1940, sect. 182). That is a strong way of saying that i-umlaut is one of the major early English vowel changes, not a marginal local effect.

Recent English-language scholarship says the same in a different register. Fulk describes the process as “**front mutation, more commonly referred to as front umlaut or i/j-umlaut**” and treats it as a large assimilatory change affecting much of the vowel system (Fulk 2018, sect. 4.7). But Fulk also gives the crucial complication for the second half of this chapter: “**diphthongization by initial palatal consonants (which precedes front umlaut but not breaking)**” (Fulk 2018, sect. 4.13).

The historical discussion therefore has two layers. The chapter center is the Old English i-umlaut itself; the West-Saxon palatal diphthongization material is real, but it stands on the edge of the chapter and has a more complicated chronological profile in the handbooks.

2. Comparison of the traditions

The traditions agree strongly on the importance of i-umlaut and less strongly on the place of the West-Saxon diphthongal material.

Campbell and Hogg both treat i-mutation as one of the central Old English vowel developments, while also recognizing a narrower palatal-vowel zone with forms such as *giefan*, *giest*, and *sceap* (Campbell 1959, sects 190–197, 248–251; Hogg 1992, 104–5, 111–14). Luick likewise gives i-Umlaut primary weight, but treats palatal diphthongization as part of a broader palatal field rather than as a second chapter of equal scale (Luick 1914–1940, sects 181–183).

Ringe and Taylor and Fulk sharpen the contrast. Both recognize the West-Saxon diphthongization material, but both place it differently in the broader textbook chronology from the local order found by the present implementation (Ringe and Taylor 2014, sects 6.5.1, 6.6.1–6.6.4; Fulk 2018, sects 4.7, 4.13). That is why the present chapter is best read as **a major umlaut chapter with a narrower West-Saxon right-edge follower**, not as a pair of fully equal historical laws.

3. Formalization in the present project

The present implementation keeps the broad umlaut rule and the narrower West-Saxon diphthongization rule separate:

```
define OEIUmlautFronting [  
  {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ā} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*á} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger  
];
```

```
define OEIUmlautRaising [  
  {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ā} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*á} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger  
];
```

```

    {*æ} -> {*e} || _ EnglishIumlautIntervening EnglishIumlautTrigger
];

define OEIumlautDiphthong [
    {*ea} -> {*ie} || _ EnglishIumlautIntervening EnglishIumlautTrigger,
    {*ēa} -> {*ie} || _ EnglishIumlautIntervening EnglishIumlautTrigger,
    {*io} -> {*ie} || _ EnglishIumlautIntervening EnglishIumlautTrigger,
    {*īo} -> {*ie} || _ EnglishIumlautIntervening EnglishIumlautTrigger,
    {*eo} -> {*ie} || _ EnglishIumlautIntervening EnglishIumlautTrigger,
    {*ēo} -> {*ie} || _ EnglishIumlautIntervening EnglishIumlautTrigger,
    {*éa} -> {*ie} || _ EnglishIumlautIntervening EnglishIumlautTrigger,
    {*éo} -> {*ie} || _ EnglishIumlautIntervening EnglishIumlautTrigger,
    {*ío} -> {*ie} || _ EnglishIumlautIntervening EnglishIumlautTrigger
];

define OEIumlaut OEIumlautFronting
    .o. OEIumlautRaising
    .o. OEIumlautDiphthong;

define OEWsPalatalDiphthongization [
    {*æ} -> {*ea} || .#. [{*tʃ} | {*dʒ} | {*f} | {*j}] _ [EnglishStarConsonant | EnglishPa
    {*ǣ} -> {*ēa} || .#. [{*tʃ} | {*dʒ} | {*f} | {*j}] _ [EnglishStarConsonant | EnglishPa
    {*e} -> {*ie} || .#. [{*tʃ} | {*dʒ} | {*f} | {*j}] _ [EnglishStarConsonant | EnglishPa
    {*ē} -> {*īe} || .#. [{*tʃ} | {*dʒ} | {*f} | {*j}] _ [EnglishStarConsonant | EnglishPa
    {*é} -> {*íe} || .#. [{*tʃ} | {*dʒ} | {*f} | {*j}] _ [EnglishStarConsonant | EnglishPa
    {*ě} -> {*īe} || .#. [{*tʃ} | {*dʒ} | {*f} | {*j}] _ [EnglishStarConsonant | EnglishPa
];

```

In prose, the distinction is straightforward. OEIumlaut performs the broad fronting, raising, and diphthongal adjustments caused by following i/j. OEWsPalatalDiphthongization is much narrower: it applies at the left edge of a word after an already palatal consonant and creates specifically West-Saxon diphthongal outputs.

4. Chronological placement

The present chronology is fixed by concrete lexical failures in both directions.

If i-umlaut is moved too early, before velar palatalization, the outputs of *cow* and *lung* become over-palatalized. The chronology card records PGmc **kūi* > *cȳ* instead of expected *cȳ*, and PGmc **lúnganjō* > *lunġen* instead of expected *lunġen* [Germanic/docs/sound_changes/order_tests/chronology_cards/SC055-oe-i-umlaut.md].

If i-umlaut is moved too late, or if West-Saxon diphthongization is moved too early, the local right edge breaks around *gift* and *sheath*. The card records PGmc **gēftiz* > *ġieft* instead of expected *ġift*, and PGmc **skáiθiz* > *scēaþ* instead of expected *scēaþ* [Ringe and Taylor (2014), §§6.5.1, 6.6.1–6.6.4; Germanic/docs/sound_changes/order_tests/chronology_cards/SC055-oe-i-umlaut.md; Germanic/docs/sound_changes/order_tests/chronology_cards/SC056-oe-ws-palatal-diphthongization.md].

The crucial methodological limit is on the later side of the West-Saxon diphthongization rule. The search found no narrower later lexical break before the present search boundary, so there is no warrant for turning that negative result into a historical claim that the rule must stand before the final stages of the cascade.

5. Consequences for reconstructed forms

The chapter changes the shape of reconstructed forms on a large scale. In the umlaut rule proper, back vowels front and high vowels raise under following i/j; this is why forms such as *cȳ* and *lunġen* belong to the same chapter as more familiar textbook examples like *giest* and *ġiefan* (Campbell 1959, sects 190–197; Fulk 2018, sect. 4.7).

The West-Saxon follower has a narrower consequence. It produces the diphthongal surface forms expected in words such as *giefan* and *sceap*, but it does so only after the broader umlautal setting is already in place (Ringe and Taylor 2014, sect. 6.5.1; Hogg 1992, 104–5). That is why the present implementation treats it as a right-edge follower rather than as a second main chapter of equal scale.

6. Remaining cautions

This chapter should not flatten two different historical scales into one.

The center is i-umlaut. The West-Saxon diphthongal material is genuine, but it is narrower, more dialect-specific, and less stably placed in the broader handbook chronology. For that reason the reader-facing chapter should keep the asymmetry visible: one major sound law with one narrower right-edge follower.

It should also avoid turning the negative later search for SC056 into a positive historical boundary. The present evidence is strong enough to fix the local relation to the left, but not to license a sweeping claim about the far right of the whole Old English sequence.

Nasal dissimilation

Internal rule and chronology card: OENasalDissimilation (SC058).

1. Historical discussion

This is not a chapter-sized textbook sound law. The strongest newer English-language formulation recovered in the local sources is Fulk’s cautious statement: “**In the cluster *mn*, the first consonant tends to lose its nasality by dissimilation, though the results are hardly regular**” (Fulk 2018, sect. 6.11). That is already a warning about scale: the phenomenon is real, but not neat or uniform.

The older German material is thinner and more lexical. Luick preserves outcomes such as *enitre* ‘*einjährig* (aus **anwintri*)’ rather than building a separate chapter around them (Luick 1914–1940). Campbell and Hogg likewise reach forms such as *heofon* in other discussions without isolating a major Old English “nasal-dissimilation” law (Campbell 1959; Hogg 1992).

The historical discussion must therefore begin by lowering expectations. The change belongs in the book because the sources preserve scattered evidence for the pattern, not because the handbooks present it as a major canonical law.

2. Comparison of the traditions

The contrast between traditions is instructive.

Newer English-language scholarship, especially Fulk, is willing to formulate a general tendency in *mn* clusters, while immediately warning that the results are irregular (Fulk 2018, sect. 6.11). Ringe and Taylor are more lexical still: their clearest direct statement is simply that *enetre* reflects “loss of the second **n* by dissimilation” (Ringe and Taylor 2014). The older German tradition, represented here by Luick, preserves useful lexical traces but does not make them into a chapter heading (Luick 1914–1940).

That comparison is the right historical scale for the present section. The rule is not fictitious, but it is better treated as a residual pattern than as a large named law comparable to *i*-umlaut or breaking.

3. Formalization in the present project

The present implementation isolates the change very narrowly:

```
define OENasalDissimilation [  
    {*m} -> {*f} || EnglishStarShortVowel _ EnglishStarShortVowel {*n} [EnglishStarShort  
];
```

The code targets medial *m* in a short-vowel environment before a following syllable containing *n*, and it rewrites that *m* as *f*. This is a stricter and more explicit formulation than the handbooks usually give. That explicitness is useful for the implementation, but it should not be mistaken for evidence that the traditional scholarship isolates exactly the same rule in exactly the same shape.

4. Chronological placement

The current chronology evidence is negative in both directions.

When the search moves the rule earlier, it finds no lexical breakpoint before the left-hand search limit at bundled *PWGmcChanges*. When it moves the rule later, it finds no lexical breakpoint before the current right-hand search limit at SC087 [*Germanic/docs/sound_changes/order_tests/chronology_cards/oe-nasal-dissimilation.md*].

That means the chapter cannot be written like the stronger pilot cases. There is no *stretch*, *cow*, or *gift*-style failure here to prove a narrow local slot. The honest statement is simpler: the search found **no lexical evidence for a narrower earlier or later boundary** within the tested window.

5. Consequences for reconstructed forms

Even so, the rule has real interpretative consequences. It provides a place in the implementation for outcomes of the *heofon*, *fæstenn*, and *enetre* type discussed in the literature (Fulk 2018, sect. 6.11; Ringe and Taylor 2014; Luick 1914–1940). Without an explicit rule, those outcomes would be left to diffuse analogy or to unexplained exception lists.

The consequence is therefore modest but real. The rule marks a narrow, partly lexicalized dissimilation tendency inside the larger Old English system. It does not reorganize the whole chronology, but it keeps a historically attested type of development visible in the model.

6. Remaining cautions

This section should stay short.

The literature does not justify treating nasal dissimilation as a chapter center of the same rank as the major textbook sound laws. Nor does the present chronology search justify a narrow slot fixed by lexical breakpoints on both sides. The right reader-facing stance is therefore deliberately modest: the rule is explicit in the present implementation, historically supported in scattered examples, and chronologically underdetermined within the tested range.

Reader-facing pilot source note

Style-model files consulted

- Germanic/docs/lexeme_reports/writing_skill/README.md
- Germanic/docs/lexeme_reports/writing_skill/book_entry_template.md
- Germanic/docs/lexeme_reports/model_entries/2183-shoulder-sculdram.model.md

Pilot source files consulted

General sound-change sources

- Germanic/docs/sound_changes/change_reports/full/052-velar-palatalization-hinge.md
- Germanic/docs/sound_changes/change_reports/full/055-056-umlaut-core.md
- Germanic/docs/sound_changes/change_reports/full/058-oe-nasal-dissimilation-residual-note.md
- Germanic/docs/sound_changes/literature_dossiers/052-velar-palatalization-hinge.dossier.md
- Germanic/docs/sound_changes/literature_dossiers/055-056-umlaut-core.dossier.md
- Germanic/docs/sound_changes/literature_dossiers/058-oe-nasal-dissimilation-residual.dossier.md
- Germanic/docs/sound_changes/order_tests/chronology_cards/SC052-oe-velar-palatalization.md
- Germanic/docs/sound_changes/order_tests/chronology_cards/SC055-oe-i-umlaut.md
- Germanic/docs/sound_changes/order_tests/chronology_cards/SC056-oe-ws-palatal-diphthongization.md
- Germanic/docs/sound_changes/order_tests/chronology_cards/SC058-oe-nasal-dissimilation.md
- Germanic/fsts/germanic.txt

Reference texts checked directly

- docs/references/campbell_old_english_grammar.txt
- docs/references/hogg_vol1.txt
- docs/references/ringe_taylor_linguistic_history_vol2.txt
- docs/references/luick_historische_grammatik.txt
- docs/references/fulk_comparative_grammar_early_germanic.vision.txt

Scope of the pilot

The pilot is intentionally small. It tests reader-facing section design, quotation method, code presentation, and chronology explanation before any full-volume rewrite is attempted.

Campbell, Alistair. 1959. *Old English Grammar*. Clarendon Press.

Fulk, R. D. 2018. *A Comparative Grammar of the Early Germanic Languages*. Vol. 3. Studies in Germanic Linguistics. John Benjamins.

Hogg, Richard M. 1992. *A Grammar of Old English. Volume 1: Phonology*. Blackwell.

Luick, Karl. 1914–1940. *Historische Grammatik Der Englischen Sprache*. Tauchnitz.

Ringe, Don, and Ann Taylor. 2014. *The Development of Old English*. Vol. 2. A Linguistic History of English. Oxford University Press.